

A Strictly Convex Counterexample for Distance-Dominance Regions

Candidate counterexample packet for arXiv:1303.4733

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Status. Candidate counterexample, likely valid, pending independent human review. The construction answers the strict-convexity conjecture in Section 5 of Reem’s paper.

1 Source problem

Daniel Reem’s paper *Topological properties of sets represented by an inequality involving distances*, arXiv:1303.4733v2 [1], proves the following for a convex subset of a uniformly convex normed space and nonempty sites P, A with $d(P, A) > 0$:

$$\partial \operatorname{dom}(P, A) = \{x : d(x, P) = d(x, A)\}, \quad \operatorname{int} \operatorname{dom}(P, A) = \{x : d(x, P) < d(x, A)\},$$

and $\operatorname{dom}(P, A)$ is the closure of the strict-inequality set, where

$$\operatorname{dom}(P, A) := \{x : d(x, P) \leq d(x, A)\}.$$

In Section 5 (PDF page 11), the paper asks whether uniform convexity can be weakened to strict convexity and states: “We conjecture that in this case there are counterexamples to Theorem 3.3.” The source passage is reproduced below.

Let $S = [-1, 1] \times \mathbb{R}$. Then $S \subset [-2, 2] \times \mathbb{R} \subseteq \operatorname{dom}(P, A)$. Thus $S \subset \operatorname{int}(\operatorname{dom}(P, A))$. But $d(z, P) = d(z, A) = d(z, (0, 0))$ for each $z \in S$. Therefore (4) does not hold.

5. CONCLUDING REMARKS

It may be of interest to further investigate the phenomenon described in this note in various domains of mathematics and to find interesting applications of it. Perhaps a discontinuous version related to the phenomenon can be formulated (a simple example where this holds: let $C = f^{\leq 0}$ where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is defined as 0 on the boundary of the unit ball, arbitrarily positive outside the ball and arbitrarily negative inside the ball). This may help in the study of singularities of the boundary.

Another possible direction for future investigation is to weaken the assumption of uniform convexity to strict convexity (the unit sphere does not contain line segments but, in contrast to uniform convexity, now there is no uniform bound $\delta(\epsilon) > 0$ on how much the midpoint $(x + y)/2$ should penetrate the unit ball assuming $|x| = |y| = 1$ and $|x - y| \geq \epsilon$). We conjecture that in this case there are counterexamples to Theorem 3.3. Alternatively, one may try to work with general normed spaces (under additional assumptions on the sites) or with spaces which are not linear. As a matter of fact, recently [39, 40] certain related results have been obtained. In the first paper (See Section 7 in the current arXiv version) a closely related result (Lemma 9.11) is used as a tool for proving the geometric stability of Voronoi cells with respect to small changes of the sites in normed spaces which are not uniformly convex, under

Figure 1: The strict-convexity conjecture in Section 5 of arXiv:1303.4733v2, PDF page 11.

2 Proof intuition

The ordinary ℓ_1 geometry makes the desired dominance relation transparent: a vector supported on two coordinates can be at least as close as a basis vector because it can absorb positive mass from either coordinate. Ordinary ℓ_1 is not strictly convex. We make it strictly convex by adding a Hilbertian term whose coordinate weights tend to zero. The perturbation enforces strict convexity globally but becomes invisible along the tail.

Take the first site to be all basis vectors and the second site to be all averages of two distinct basis vectors. These sites remain one unit apart in the ℓ_1 part of the norm. Their distances from the origin both have infimum one, attained only asymptotically. For every point in a fixed ball, a second averaging coordinate can be sent far enough into the tail that the Hilbertian cost vanishes; the remaining coordinate then gives a strictly smaller Hilbertian contribution than the corresponding basis vector. Thus the whole ball lies in the dominance region although the origin is an equality point.

3 Counterexample

Let $X = \ell_1(\mathbb{N}; \mathbb{R})$, put $w_n = 2^{-n}$, and define

$$q(x) := \left(\sum_{n=1}^{\infty} w_n x_n^2 \right)^{1/2}, \quad \|x\|_* := \|x\|_1 + q(x).$$

Write (e_n) for the standard unit vectors and set

$$A := \{e_n : n \in \mathbb{N}\}, \quad P := \left\{ \frac{e_n + e_m}{2} : n, m \in \mathbb{N}, n \neq m \right\}.$$

Theorem 1. *The norm $\|\cdot\|_*$ makes X a strictly convex Banach space. The sets P and A are closed and satisfy*

$$d_*(P, A) = 1, \quad d_*(0, P) = d_*(0, A) = 1.$$

Moreover,

$$B_*(0, 1/2) \subseteq \text{dom}(P, A).$$

Consequently, 0 is an interior point of $\text{dom}(P, A)$ at which $d_*(0, P) = d_*(0, A)$. Hence

$$\partial \text{dom}(P, A) \neq \{x : d_*(x, P) = d_*(x, A)\}$$

and

$$\text{int dom}(P, A) \neq \{x : d_*(x, P) < d_*(x, A)\}.$$

In particular, strict convexity cannot replace uniform convexity in Theorem 3.3 of [1] for arbitrary positively separated sites.

Proof. Since $q(x) \leq \|x\|_2 \leq \|x\|_1$,

$$\|x\|_1 \leq \|x\|_* \leq 2\|x\|_1.$$

Thus $\|\cdot\|_*$ is an equivalent Banach norm on ℓ_1 . The norm q is a strictly convex norm because every w_n is positive. If equality held in the midpoint inequality for $\|\cdot\|_*$ at two distinct unit vectors, equality would have to hold for both the ℓ_1 and q summands. Equality for q forces the two vectors

to be nonnegative scalar multiples of one another, and equality of their $\|\cdot\|_*$ -norms then forces them to coincide. Hence $\|\cdot\|_*$ is strictly convex.

For $p_{n,m} := (e_n + e_m)/2$ and $e_k \in A$,

$$\|p_{n,m} - e_k\|_1 = \begin{cases} 1, & k \in \{n, m\}, \\ 2, & k \notin \{n, m\}. \end{cases}$$

Therefore $d_*(P, A) \geq 1$. Sending two distinct indices to infinity gives

$$\|p_{n,m} - e_n\|_* = 1 + \frac{1}{2}\sqrt{w_n + w_m} \longrightarrow 1,$$

so $d_*(P, A) = 1$. Distinct elements of A are at ℓ_1 -distance 2, and distinct elements of P (viewed as unordered pairs) are at ℓ_1 -distance at least 1. Thus both sites are uniformly discrete and hence closed.

At the origin,

$$\|e_n\|_* = 1 + \sqrt{w_n} \longrightarrow 1$$

and, as $n, m \rightarrow \infty$ with $n \neq m$,

$$\|p_{n,m}\|_* = 1 + \frac{1}{2}\sqrt{w_n + w_m} \longrightarrow 1.$$

All these vectors have ℓ_1 -norm one, so both infima equal one:

$$d_*(0, A) = d_*(0, P) = 1.$$

It remains to prove the ball inclusion. Fix $x \in X$ with $\|x\|_* < 1/2$. Then $|x_j| < 1/2$ for every j . Let $x_j^+ = \max\{x_j, 0\}$. For every fixed n ,

$$\|x - e_n\|_1 = \|x\|_1 + 1 - 2x_n^+, \quad (1)$$

$$q(x - e_n)^2 = q(x)^2 + w_n(1 - 2x_n). \quad (2)$$

Choose indices $m \rightarrow \infty$ with $m \neq n$. Since $x \in \ell_1$, we have $x_m \rightarrow 0$, and of course $w_m \rightarrow 0$. The same coordinate calculation gives

$$\begin{aligned} \|x - p_{n,m}\|_1 &= \|x\|_1 + 1 - 2x_n^+ - 2x_m^+, \\ q(x - p_{n,m})^2 &= q(x)^2 + w_n(1/4 - x_n) + w_m(1/4 - x_m). \end{aligned}$$

Consequently,

$$\begin{aligned} \lim_{m \rightarrow \infty} \|x - p_{n,m}\|_* &= \|x\|_1 + 1 - 2x_n^+ + \sqrt{q(x)^2 + w_n(1/4 - x_n)}. \end{aligned} \quad (3)$$

Because $x_n < 1/2 < 3/4$,

$$q(x)^2 + w_n(1/4 - x_n) < q(x)^2 + w_n(1 - 2x_n).$$

Equations (1)–(3) therefore imply

$$d_*(x, P) \leq \lim_{m \rightarrow \infty} \|x - p_{n,m}\|_* < \|x - e_n\|_* \quad (n \in \mathbb{N}).$$

In particular, $d_*(x, P) \leq \inf_n \|x - e_n\|_* = d_*(x, A)$. Hence $x \in \text{dom}(P, A)$, proving $B_*(0, 1/2) \subseteq \text{dom}(P, A)$.

The origin is therefore an interior point of the dominance region, but its distances to the two sites are equal. It cannot be a boundary point, and both asserted failures follow. $\square$

4 Verification notes

The proof uses only coordinate identities. The main points for independent review are:

1. positivity of every w_n , which makes the Hilbertian summand strictly convex;
2. the exact ℓ_1 separation of the two sites;
3. nonattainment of both distance-one infima at the origin;
4. the passage $m \rightarrow \infty$ in (3); and
5. the radicand gap $w_n(3/4 - x_n) > 0$ on the stated ball.

A deterministic numerical checker in `code/verifier.py` evaluates these identities on 500 seeded random finite-support vectors and tests midpoint strictness on a further 500 random pairs. It is a sanity check, not part of the proof.

5 Novelty and limitations

This packet gives a counterexample to the strict-convexity conjecture, not a characterization of all strictly convex spaces or all site configurations for which Reem’s conclusions remain true. The use of nonattained site distances is essential to this construction and matches the infinite-dimensional case explicitly left outside the auxiliary results discussed by the source paper.

The bounded novelty check performed on 9 August 2026 searched the run registry and attempts for arXiv:1303.4733 and the exact title, and searched arXiv/web indexes with the phrases “positively separated sites strictly convex Voronoi”, “fat bisector strictly convex Banach space”, “uniform convexity to strict convexity Voronoi counterexample”, and the source title together with “strict convexity”. The search returned the source paper and general finite-dimensional bisector literature, but no later paper explicitly resolving this conjecture and no occurrence of this construction. This is not an exhaustive citation review; novelty confidence is therefore moderate, while mathematical confidence in the self-contained construction is high.

Human-review recommendation. Review as a likely full counterexample. The decisive check is the pointwise comparison between (2) and (3); after that, verify that taking an infimum over n preserves the dominance inequality.

References

- [1] D. Reem, *Topological properties of sets represented by an inequality involving distances*, arXiv:1303.4733v2 (2013), Section 5, p. 11. <https://arxiv.org/abs/1303.4733>.